

ELIAS SAVAGE

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OVERVIEW

Second-year Product Design Engineering (BEng) student at Loughborough University with a 76% first-year average. Hands-on engineer with experience taking ideas from concept to working prototype, combining Siemens NX CAD, sensor-driven electronics, engineering analysis and workshop skills. Seeking a year-long industrial placement in product design engineering to apply technical knowledge to real-world products.

EDUCATION

BEng Product Design Engineering, Loughborough University 2025 – 2029

- Year 1 average: 76% (First-class standard)
- Top modules: Engineering Science (86%), Integrated Product Design (81%), Applied Engineering (80%)
- Also studied: Mathematics for Product Design Engineering (74%), Product Design Methods (68%), Materials and Manufacturing Processes (65%)

A Levels, Product Design (A*), Mathematics (B), Physics (B), Godalming College 2025
GCSEs, 9 GCSEs (grades 9–5) including Maths (8), Design Technology (8), Geography (9), Computer Science (7), Science (7, 7), King Edward's School Witley 2023

EXPERIENCE

EWB Smart Plant Pot 2025
Loughborough University, Group Project *Loughborough, UK*

- Designed a smart plant pot with a six-person team for an Engineers Without Borders brief, helping children in Ladywood, Birmingham to grow vegetables
- Prototyped and tested four 3D-printed LED geometries, measuring lux at set distances to identify the dome as the most efficient shape. We found a dome provided 45% more light than a flat ring with the same light source.
- Specified a PID-controlled grow light, using PPFD and DLI calculations to validate the LED choice
- Modelled almost all of the 11-part assembly in Siemens NX

BRE Damp Prevention System 2025
Loughborough University, Group Project *Loughborough, UK*

- Led system ideation for a three-person team on a Building Research Establishment brief, ranking features and defining measurable requirements
- Built and tested an Arduino prototype combining humidity and motion sensors, a servo-driven extractor fan and an RGB LCD display
- Evaluated real-world performance critically, documenting bugs and gaps between test and deployment conditions

AeroPress CAD Visualisation 2026
Loughborough University *Loughborough, UK*

- Reverse-engineered an AeroPress into a dimensionally accurate multi-part CAD model in Siemens NX
- Produced ray-traced renders across four scene types, aligning HDR environments for accurate lighting and reflections

BS8888 Engineering Drawings 2025
Loughborough University *Loughborough, UK*

- Produced a toleranced part drawing with section views and hole callouts to BS8888

- Created a full assembly documentation set for a stepper-motor linear stage, including sectioned, exploded and isometric views with a parts list

Modular Wall Light

A Level Product Design, Godalming College

2024

Godalming, UK

- Designed and built a dimmable wall light with interchangeable laser-cut acrylic shades, earning an A* grade
- Integrated 3D-printed fixings, timber and a 24V COB LED system with remote dimming

Design and Construction

Self-Employed

August 2023

Guildford, UK

- Designed a bespoke seating area for a client's asymmetric garden space, balancing function and feasibility
- Produced full project costings and material lists
- Built the design alongside a local tradesperson, repairing existing decking in the process

SKILLS

CAD & Visualisation

Siemens NX (modelling, assemblies, rendering), BS8888 drawing

Prototyping

3D printing, laser cutting, CNC machining, woodworking, bent lamination

Electronics

Arduino, sensor integration, LED systems, low-voltage wiring

Analysis

Testing and validation, PID control, light and data analysis

EXTRA-CURRICULAR

Skateboarding. Skateboarding since 2019 and involved in the local skate community, including featuring in an advert for a local skate shop. Skateboarding has taught me that success is achieved through failure, an important lesson I find applies in many other situations. Building half pipe ramps for skate boarding is also a hobby.

Weighted Calisthenics and Gym. Training with structured, progressive programmes has taught me a systematic and calculated approach to tasks, breaking long-term goals into measurable steps.

Field Hockey. Played field hockey throughout school and college, including for Godalming College's men's first team.